

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No .....

## **CS-5005(3)-CBGS**

### **B.E. V Semester**

Examination, December 2020

### **Choice Based Grading System (CBGS)**

### **Artificial Intelligence**

*Time : Three Hours*

*Maximum Marks : 70*

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the state space with the use of Water jug problem.  
b) Explain Production system and production characteristics,
2. a) What are the characteristics of good knowledge representation technique? Explain each of them in brief.  
b) Explain how conventional computing differ from the Intelligence Computing.
3. a) Explain the Best First Search Procedure with example.  
b) Construct the predicate logic for the following
  - i) John likes all kind of Food
  - ii) Apples are Food
  - iii) Chicken is Food
4. a) Explain Semantic Nets and Frame. Construct semantic net representation for the following:  
“Mary gave the green flowered vase to her favorite cousin”.  
b) What is alpha-beta pruning? Discuss it with example.

CS-5005(3)-CBGS

PTO

[2]

5.
  - a) What do you mean by Game Playing? Explain with algorithm and example: Minimax algorithm.
  - b) Define Natural Language generation. List any two NLP systems. What is the role of semantic analysis in NLP?
6.
  - a) Write down the minimax algorithm and explain with the help of an example.
  - b) What are the component of a planning system?
7.
  - a) Discuss Forward and Backward chaining.
  - b) Outline the main steps of the knowledge engineering process in detail.
8. Write short notes (any three)
  - a) MYCIN
  - b) AO\* algorithm
  - c) Frames
  - d) Application of neural network
  - e) Hill climbing

\*\*\*\*\*